

Continue Minor Project (Malware Detection Tool)

Implement basic scanning functionality in your Python-based malware detection tool.

Try to detect malicious files based on hash signatures.

Submission: document everything and push it as task 3

Upgradation in previous script file :

```
(hacker@kali)-[~/MalwareScanner]
$ python3 malware_scanner.py

Scanning directory: ./test_files

[CHECKING] test.txt => d2a84f4b8b650937ec8f73cd8be2c74add5a911ba64df27458ed8229d
a804a26
✅ test.txt is safe.
[CHECKING] ls => 54f2c54357c31dac1ec5adc40b051744ea03a57ad40427cb6a2a5f7cfe1c0ee
b
✅ ls is safe.
[CHECKING] sample.txt => c98c24b677eff44860afea6f493bbaec5bb1c4cbb209c6fc2bbb47f
66ff2ad31
✅ sample.txt is safe.
```

Updated Python script :

```
import os
```

```
import hashlib
```

```
# --- CONFIGURATION ---
```

```
MALWARE_HASHES = {
```

```
    "44d88612fea8a8f36de82e1278abb02f", # Example malware hash
```

```
    "098f6bcd4621d373cade4e832627b4f6"
```

```
}
```

```
SCAN_DIRECTORY = "./test_files" # Directory to scan
```

--- FUNCTION: Compute SHA-256 Hash ---

```
def compute_sha256(file_path):  
    hasher = hashlib.sha256()  
  
    try:  
        with open(file_path, "rb") as f:  
            for chunk in iter(lambda: f.read(4096), b''):  
                hasher.update(chunk)  
        return hasher.hexdigest()  
  
    except Exception as e:  
        print(f"Error hashing {file_path}: {e}")  
        return None
```

--- FUNCTION: Scan Directory ---

```
def scan_directory():  
  
    print(f"\n🔍 Scanning directory: {SCAN_DIRECTORY}\n")  
  
    for root, _, files in os.walk(SCAN_DIRECTORY):  
        for file in files:  
            file_path = os.path.join(root, file)  
            file_hash = compute_sha256(file_path)  
  
            if file_hash:  
                print(f"[CHECKING] {file} => {file_hash}")  
  
            # Compare hash with known malware hashes  
  
            if file_hash in MALWARE_HASHES:  
                print(f"⚠️ [WARNING] Malicious file detected: {file} ⚠️ ")
```

else:

print(f"✅ {file} is safe.")

Run the scan

if __name__ == "__main__":

scan_directory()